

Case study Solution

Q. Book My Show, India's top entertainment platform faced inefficiencies due to over provisioned on-premises servers and struggled with scalability during peak traffic. How to ensure optimization of infrastructure, cost, scalability & Performance. Solve the above case study with appropriate explanation.

Ans.

Problem Statement :

Bookmyshow faced :

- over-provisioned on-premises servers -
High - Infrastructure cost and resource wastage
- Scalability during peak traffic -
System slowdown or crashes during New movie release
- Performance Issue -
Poor user experience during high demand.

Solutions :

To solve above problems BookMyshow can adopt cloud computing with following strategies

① Move (migrate) to cloud infrastructure -

① Move application from on-premises servers to cloud platforms like AWS, Azure, Google cloud

② cloud provides on-demand resources, so no need to maintain extra hardware

Result : Reduce Infrastructure cost & eliminates Over-provisioning

② Auto - Scaling -

① Auto-scaling to automatically increases or decreases resources based on traffic

② During peak time (movie release), servers scale up; during low traffic, scale down

Result: Increase Scalability & Reduce downtime

③ Cost Optimization -

① Use pay-as-you-go Pricing model

② Implement reserved instances or spot instances for predictable workloads

Result: Pay only for what is used, reducing unnecessary expenses

④ Load Balancing -

① Distribute traffic across multiple cloud or multiple server using load balancers

Result: Prevents server overload & improves system performance

⑤ Content Delivery Network (CDN)

① Use CDNs (like CloudFront) to deliver content from servers closer to users

Result: Faster response time & better user experience

⑥ Database Optimization -

① Use managed databases (eg. AWS RDS, DynamoDB)

② Implement caching to reduce database load

Result: Improves speed and handles high traffic efficiently.

⑦ Microservices Architecture -

- ① Break application into smaller services (Booking, payment, search)

Result: Improves Scalability & Fault isolation

⑧ Monitoring & Performance Management

- ① Use Tools like cloudwatch, Azure Monitor

Result: Real-time monitoring & quick issue resolution

Final Outcome:

By adopting these strategies, BookMyShow can achieve following things.

- ① Optimized infrastructure usage
- ② Reduced operational cost
- ③ High scalability during peak traffic
- ④ Improved performance and user experience.